Create new polyline layer for KML mission path without segmented sections

Unsplit Line (Data Management Tools)

- Input Features: KMLpolylines_STX2025
- Output Feature Class: KMLSTX2025_PATHLINE
- Dissolve Fields: Leave empty (dissolve all fields to create single line)

BACKGROUND DEPTH & SLOPE CONDITIONS ----------

NOTE: Using GEBCO 15 arc-second resolution rasters for background calculations due to extremely long loading time for Jeremiah's 10-meter resolution rasters

Create buffer around KML mission path line that background rasters will be cropped to in order to accurately reflect mission conditions (instead of conditions in entire raster clip for map)

Pairwise Buffer (Analysis Tools):

- Input Features: KMLSTX2025_PATHLINE
- Output Feature Class: C:\Users\kkeii\OneDrive\Documents\ArcGIS\Files\THESIS_Files\Missions\mission_UVI2_Around_STX_20250306\mission_UVI2_Around_STX_20250306.gdb\KMLSTX2025_PATHLINE_BUFFER
- Distance [value or field]: 1500 Linear Unit: Meters
- Method: Planar
- Dissolve Type: Dissolve all output features into a single feature

Add mission GEBCO 15 arc-sec bathymetry (depth) raster without shading for calculations

+ Add to Map GBC15a_bathy_STX2025 layer WITHOUT shaded relief

Clip bathy (depth) raster to buffer around KML mission path line

Extract by Mask (Spatial Analyst Tools):

- Input Raster: GBC15a_bathy_STX2025
- Input Raster or Feature Mask Data: KMLSTX2025_PATHLINE_BUFFER
- Output Raster: C:\Users\kkeii\OneDrive\Documents\ArcGIS\Files\THESIS_Files\Missions\mission_UVI2_Around_STX_20250306\mission_UVI2_Around_STX_20250306.gdb\GBC15a_bathyBACKGROUND_STX2025
- Extraction Area: Inside (inside = keep the raster cells that fall inside your mask polygon)
- Environments > Cell Size: Same as layer GBC15a_bathy_STX2025
- Environments > Snap Raster: GBC15a_bathy_STX2025 (ensures perfect cell alignment)
- Environments > Compression: NONE

Repeat for slope raster

Extract by Mask (Spatial Analyst Tools):

- Input Raster: GBC15a_slope_STX2025
- Input Raster or Feature Mask Data: KMLSTX2025_PATHLINE_BUFFER

- Output Raster: C:\Users\kkeii\OneDrive\Documents\ArcGIS\Files\THESIS_Files\Missions\mission_UVI2_Around_STX_20250306\mission_UVI2_Around_STX_20250306.gdb\GBC15a_slopeBACKGROUND_STX2025
- Extraction Area: Inside (inside = keep the raster cells that fall inside your mask polygon)
- Environments > Cell Size: Same as layer GBC15a_slope_STX2025
- Environments > Snap Raster: GBC15a_slope_STX2025 (ensures perfect cell alignment)
- Environments > Compression: NONE

*Convert background bathy (depth) raster to point feature class layer for calculations ('**grid_code**' column has actual bathymetric depth values!)*

Raster to Point (Conversion Tools)

- Input Raster: GBC15a_bathyBACKGROUND_STX2025
- Field: Value
- Output Point Features: GBC15a_bathyPOINTS_STX2025

*Repeat for slope raster ('**grid_code**' column has actual bathymetric slope values!)*

Raster to Point (Conversion Tools)

- Input Raster: GBC15a_slopeBACKGROUND_STX2025
- Field: Value
- Output Point Features: GBC15a_slopePOINTS_STX2025

Add columns for x- and y-coordinates (in decimal degrees) to attribute table for new bathy depth point layer (so that data for the location of each point is preserved if the attribute table is exported as a CSV)

Calculate Geometry Attributes (Data Management Tools)

- Input Features: GBC15a_bathyPOINTS_STX2025
- Geometry Attributes >

Fields (Existing or New):	Property:
latitude_DD	Point y-coordinate
longitude_DD	Point x-coordinate

- Coordinate Format: Decimal Degrees
- Coordinate System: Leave blank (defaults to input features coordinate system NAD 1983 UTM Zone 20N / VIVD09 height)

Repeat for slope

Calculate Geometry Attributes (Data Management Tools)

- Input Features: GBC15a_slopePOINTS_STX2025
- Geometry Attributes >

Fields (Existing or New):	Property:
latitude_DD	Point y-coordinate
longitude_DD	Point x-coordinate

- Coordinate Format: Decimal Degrees
- Coordinate System: Leave blank (defaults to input features coordinate system NAD 1983 UTM Zone 20N / VIVD09 height)

Export bathy raster point layer WITH added coordinates as feature class to mission geodatabase for safekeeping

Right-click layer in Map Contents pane GBC15a_bathyPOINTS_STX2025 **-> Data -> Export Features:**

- Input Features: GBC15a_bathyPOINTS_STX2025
- Output Feature Class: C:\Users\kkeii\OneDrive\Documents\ArcGIS\Files\THESIS_Files\Missions\mission_UVI2_Around_STX_20250306\mission_UVI2_Around_STX_20250306.gdb\background_bathydepth_GBC15a_STX2025

*Export CSV of data in bathy raster point layer for calculations of depth background conditions (*must save **OUTSIDE OF A GDB** to get CSV format)*

Right-click layer in Map Contents pane GBC15a_bathyPOINTS_STX2025 **-> Data -> Export Table:**

- Input Table: GBC15a_bathyPOINTS_STX2025
- Output Table: C:\Users\kkeii\OneDrive\Documents\ArcGIS\Files\THESIS_Files\Missions\mission_UVI2_Around_STX_20250306\Output\background_bathydepth_GBC15a_STX2025.csv

Repeat for slope point feature class

Right-click layer in Map Contents pane GBC15a_slopePOINTS_STX2025 **-> Data -> Export Features:**

- Input Features: GBC15a_slopePOINTS_STX2025
- Output Feature Class: C:\Users\kkeii\OneDrive\Documents\ArcGIS\Files\THESIS_Files\Missions\mission_UVI2_Around_STX_20250306\mission_UVI2_Around_STX_20250306.gdb\background_slope_GBC15a_STX2025

*Repeat for slope CSV (*must save **OUTSIDE OF A GDB** to get CSV format)*

Right-click layer in Map Contents pane GBC15a_slopePOINTS_STX2025 **-> Data -> Export Table:**

- Input Table: GBC15a_slopePOINTS_STX2025
- Output Table: C:\Users\kkeii\OneDrive\Documents\ArcGIS\Files\THESIS_Files\Missions\mission_UVI2_Around_STX_20250306\Output\background_slope_GBC15a_STX2025.csv

BACKGROUND DISTANCE TO SHORELINE ----------

Create point feature class layer from reprojected KML line that consists of points every 100 meters along mission path for calculations

Generate Points Along Lines (Data Management Tools)

- Input Features: KMLSTX2025_PATHLINE
- Output Feature Class: KMLSTX2025_BKGPOINTS
- Point Placement: By Distance

- Distance: 100 Meters
- ✅ Add accumulated distance and sequence fields
- Distance Method: Planar

Rename columns in new KMLSTX2025_BKGPOINTS layer Attribute Table

Attribute Table KMLSTX2025_BKGPOINTS -> **Fields Editor** -> rename columns:

- ORIG_LEN -> dist_from_pt1_m (cumulative distance of each point from first point/row)
- ORIG_SEQ -> pointID
- Shape_Length -> length_entire_line_m

Calculate distance from each point to nearest shoreline in entire CARIBBEAN_SHORELINES layer

Near (Analysis Tools)

- Input Features: KMLSTX2025_BKGPOINTS
- Near Features: CARIBBEAN_SHORELINES
- Search Radius: Leave blank (and leave default Meters)
- ✅ Location (adds NEAR_X and NEAR_Y columns with x- and y-coordinates of nearest shoreline)
- Method: Planar
- Field Names: Leave as default EXCEPT change NEAR_DIST to NEAR_DIST_m
- Distance Unit: Meters

Create and populate 'nearest_shoreline' column in KMLSTX2025_POINTS Attribute Table with full names of nearest shoreline using numeric codes in NEAR_FID column

Attribute Table CARIBBEAN_SHORELINES -> **make note of names** for rows where 'OBJECTID' column matches numbers in KMLSTX2025_BKGPOINTS 'NEAR_FID' column

Calculate Field (Data Management Tools):

- Input Table: KMLSTX2025_BKGPOINTS
- Field Name (Existing or New): nearest_shoreline
- nearest_shoreline =

```
get_name(!NEAR_FID!)
```

- Code Block:

```
def get_name(fid):
    if fid == 847:
        return "Saint Croix, U.S. Virgin Islands"
    elif fid == 850:
        return "Buck Island, , U.S. Virgin Islands"
    < etc… >
    else:
        return Non
```

Export KML point layer WITH added NEAR_DIST columns as feature class to mission geodatabase for safekeeping

Right-click layer in Map Contents pane KMLSTX2025_BKGPOINTS **-> Data -> Export Features:**

- Input Features: KMLSTX2025_BKGPOINTS
- Output Feature Class: C:\Users\kkeii\OneDrive\Documents\ArcGIS\Files\THESIS_Files\Missions\mission_UVI2_Around_STX_20250306\mission_UVI2_Around_STX_20250306.gdb\background_distshore_STX2025

*Export CSV of data in KML point layer for nearest shoreline calculations of background conditions (*must save **OUTSIDE OF A GDB** to get CSV format)*

Right-click layer in Map Contents pane KMLSTX2025_BKGPOINTS **-> Data -> Export Table:**

- Input Table: KMLSTX2025_BKGPOINTS
- Output Table: C:\Users\kkeii\OneDrive\Documents\ArcGIS\Files\THESIS_Files\Missions\mission_UVI2_Around_STX_20250306\Output\background_distshore_STX2025.csv
-
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- MLSTX2025_BKGPOINTS.csv (*must save OUTSIDE of a gdb to get CSV format)